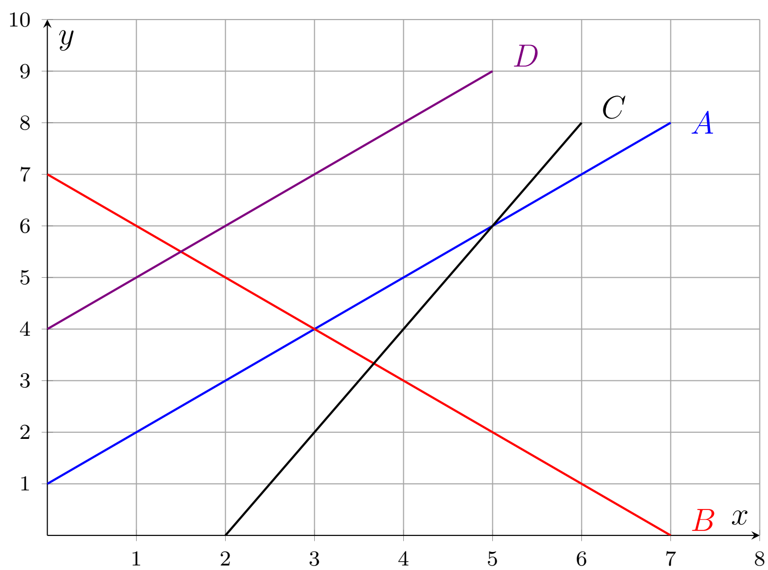
Where Two Lines Meet

The point that makes both equations true at once

Directions

- A *solution of a system* is a pair of values that makes *every* equation true — not just one of them.
- On a graph that pair is the point where the two lines cross. In a table it is the row the two tables share.
- Write every answer as an ordered pair, and say what the two numbers stand for when the problem comes from a story.

Graph Bank



Line A is $y = x + 1$, line B is $y = 7 - x$, line C is $y = 2x - 4$, and line D is $y = x + 4$. Only the problems that name a line by letter use this graph.

1. Find the pair of values that makes *both* of these equations true: $x + y = 5$ and $x - y = 3$. Write your answer as an ordered pair.

Answer: ________________

2. Find the pair of values that makes both of these equations true: $2x + y = 11$ and $x - y = 1$. Write your answer as an ordered pair.

Answer: ____________________

3. Lines A and B in the Graph Bank are the graphs of $y = x + 1$ and $y = 7 - x$.

(a) Write the coordinates of the point where the two lines cross.

Answer: ____________________

(b) Explain what that point means for the two equations.

Answer: ____________________

4. Two bike-rental plans are shown in the tables below. Plan A follows $y = 4h + 8$ dollars and plan B follows $y = 2h + 16$ dollars, where h is the number of hours.

Plan A hours	1	2	3	4
Plan A cost (\$)	12	16	20	24

Plan B hours	1	2	3	4
Plan B cost (\$)	18	20	22	24

(a) What does plan A cost at 4 hours?

Answer: ________________

(b) What does plan B cost at 4 hours?

Answer: ________________

(c) Write the (hours, cost) pair the two tables share — the solution of the system.

Answer: ________________

5. Gym P charges \$25 to join plus \$8 for each class. Gym Q charges \$40 to join plus \$8 for each class. Their cost equations form a system.

(a) Find the number of classes at which the two gyms cost the same, or state that there is no such number.

Answer: ________________

(b) Explain what your answer means about the two gyms.

Answer: ________________

6. Lines A and C in the Graph Bank are the graphs of $y = x + 1$ and $y = 2x - 4$. Write the coordinates of the point where they cross.

Answer: ________________

7. A student claims that $(3, 2)$ is the solution of the system $3x + y = 11$ and $x + 2y = 12$.

(a) Find the actual solution of the system.

Answer: ________________

(b) Explain what happens when the claimed pair is substituted into each equation, and why that settles the question.

Answer: ________________

8. Plan J's data charges are in the table below; its equation is $y = 6x + 20$ dollars for x gigabytes. Plan K starts at $32 and follows $y = 2x + 32$ dollars.

Gigabytes	0	1	2	3
Plan J cost ($)	20	26	32	38

(a) What does plan J cost at 3 gigabytes?

Answer: ________________

(b) Write the (gigabytes, cost) pair at which the two plans agree.

Answer: ________________

9. Lines A and D in the Graph Bank are the graphs of $y = x + 1$ and $y = x + 4$.

(a) Write the solution of this system, or state that there is none.

Answer: ________________

(b) Explain how the Graph Bank shows that your answer to part (a) is right.

Answer: ________________

10. A community pool sells single visits for \$5 each, so v visits cost $y = 5v$ dollars. A membership costs \$30 plus \$2 for each visit, so it costs $y = 2v + 30$ dollars.

(a) Write the (visits, cost) pair at which the two options cost the same.

Answer: __________________

(b) Explain what that pair means for a swimmer deciding between the two options.

Answer: __________________